Lecture 4: Subspace and Linear Independence

Proving Rank is Well-Defined Through Geometry

Motivation: Is Rank Well-Defined?

Recall: Rank from Cross-Filling (Lecture 3)

In Lecture 3, we defined:

Definition (Rank — Algorithmic Definition)

$$\text{rank}(A) = \text{number of rank-one pieces from cross-filling}$$

Cross-filling decomposes:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where each R_i is rank-one, and r is the rank.

The Problem: Different Pivot Choices

Cross-filling allows **different pivot choices** at each step.

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

We could pick pivot at:

- Position $(1,1)$ with value 2
- Position $(2,2)$ with value 2
- Position $(3,3)$ with value 9

Question: Do all choices give the **same number** of pieces?

Today's Goal

Prove that rank is well-defined by discovering:

$$\boxed{\text{rank}(A) = \text{dimension of column space}}$$

Strategy:

1. Understand **two languages** for describing sets
2. Define **subspaces** and **column space**
3. Introduce **linear independence**, **basis**, **dimension**
4. Prove cross-filling produces a **basis** for column space
5. Prove all bases have **same size** (using trace)
6. Conclude: rank = dimension (well-defined!)

We elevate rank from **algorithm** to **geometry**.

Two Languages for Describing Sets

Daily Life: Finding Your Classroom

Suppose you want to tell someone **where your classroom is**.

Method A (Descriptive): List properties it satisfies

- "Find the room that has exactly 30 desks"
- "Has windows facing south"
- "Is on the 2nd floor"

Method B (Constructive): Give directions to generate it

- "Go to 2nd floor"
- "Walk to end of hall"
- "Turn left, last door"

Two Languages: Comparison

	Descriptive	**Constructive**
How	List **properties** members must satisfy	Provide **recipe** to generate members
Strength	Easy to **verify** membership	Easy to **generate** new members
Weakness	Hard to **produce** members	Hard to **verify** membership
Example	"Red hat wearers"	"2nd floor, last door left"

Key: Neither is better — they're **complementary**! We need both.

Example: Line Through Origin in $\mathbb{R}^2$

Consider the line $y = 2x$:

Descriptive language (equation):

$$L = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : y = 2x \right\}$$

- To **verify** $\begin{pmatrix} 3 \\ 6 \end{pmatrix} \in L$: Check $6 = 2(3)$ ✓
- To **generate** members: Hard! Need to solve equation

Constructive language (parametric):

$$L = \left\{ t \begin{pmatrix} 1 \\ 2 \end{pmatrix} : t \in \mathbb{R} \right\}$$

- To **generate** members: Pick any t! E.g., $t = 3$ gives $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$
- To **verify** $\begin{pmatrix} 3 \\ 6 \end{pmatrix} \in L$: Hard! Find t such that $t \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

Example: Plane Through Origin in $\mathbb{R}^3$

Descriptive:

$$P = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x + y + z = 0 \right\}$$

Easy to verify: Check if sum $= 0$

Constructive:

$$P = \left\{ s \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} : s, t \in \mathbb{R} \right\}$$

Easy to generate: Pick any s, t

Same plane, two languages!

Connection to Matrices

For a matrix A:

Null space (descriptive):

$$\{x : Ax = 0\}$$

Described by an **equation** that members must satisfy.

Column space (constructive):

$$\{Ax : x \in \mathbb{R}^n\}$$

Described by a **recipe** to generate members.

Solving Equations Overcomes Language Weaknesses

Constructive language weakness: Hard to **verify** if $b \in \text{Col}(A)$

Given b, is it a linear combination of columns of A? Hard to check directly!

How to compensate? Switch to the **descriptive/equation perspective**:

$$\text{Solve } Ax = b$$

- If solution exists $\rightarrow b \in \text{Col}(A)$ ✓
- If no solution $\rightarrow b \notin \text{Col}(A)$ $\times$

Key: **Solving equations** is the **tool** to overcome constructive language's verification weakness!

Not "understanding language helps solve" — it's "solving helps overcome language limitation".

Subspaces: Flat Spaces Through Origin

Definition: Subspace

Definition (Subspace)

A **subspace** W of $\mathbb{R}^n$ is a non-empty subset that is **closed under linear combinations**:

For any $\mathbf{u}, \mathbf{v} \in W$ and any scalars $\alpha, \beta \in \mathbb{R}$:

$$\alpha\mathbf{u} + \beta\mathbf{v} \in W$$

Equivalently, check these three conditions:

1. $\mathbf{0} \in W$ (contains zero vector)
2. If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$ (closed under addition)
3. If $\mathbf{u} \in W$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{u} \in W$ (closed under scaling)

Geometric Intuition: Flat Spaces

A subspace is a **flat space** that:

- Passes through the **origin** (must contain $\mathbf{0}$)
- Is **closed** under vector operations

Examples of subspaces:

- Lines through origin in $\mathbb{R}^2$ or $\mathbb{R}^3$
- Planes through origin in $\mathbb{R}^3$
- All of $\mathbb{R}^n$ itself
- Just $\{\mathbf{0}\}$ (the zero subspace)

Non-examples:

- Line $y = 2x + 1$ (doesn't pass through origin)
- Unit circle (not closed under scaling: $2 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ not on circle)

Example: Verifying a Subspace

Question: Is $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : x + 2y - z = 0 \right\}$ a subspace of $\mathbb{R}^3$?

Step 1: Check $\mathbf{0} \in W$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} : \quad 0 + 2(0) - 0 = 0 \quad \checkmark$$

Step 2: Check closure under linear combinations

Take any $\mathbf{u}, \mathbf{v} \in W$. Then:

- $u_1 + 2u_2 - u_3 = 0$ (since $\mathbf{u} \in W$)
- $v_1 + 2v_2 - v_3 = 0$ (since $\mathbf{v} \in W$)

Example: Verifying a Subspace (continued)

For any scalars α, β, check $\alpha\mathbf{u} + \beta\mathbf{v} \in W$:

$$\alpha\mathbf{u} + \beta\mathbf{v} = \begin{pmatrix} \alpha u_1 + \beta v_1 \\ \alpha u_2 + \beta v_2 \\ \alpha u_3 + \beta v_3 \end{pmatrix}$$

Verify the equation:

$$\begin{aligned} &(\alpha u_1 + \beta v_1) + 2(\alpha u_2 + \beta v_2) - (\alpha u_3 + \beta v_3) \\ &= \alpha(u_1 + 2u_2 - u_3) + \beta(v_1 + 2v_2 - v_3) \\ &= \alpha(0) + \beta(0) = 0 \quad \checkmark \end{aligned}$$

Conclusion: W is a subspace! (It's a plane through origin)

Non-Example: Line Not Through Origin

Question: Is $U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x + y = 1 \right\}$ a subspace?

Quick check: Is $\mathbf{0} \in U$?

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} : \quad 0 + 0 = 0 \neq 1 \quad \times$$

Conclusion: NOT a subspace!

This is a line not through the origin.

Column Space: The Constructive Subspace

Definition: Column Space

Definition (Column Space)

For an $m \times n$ matrix $A = \begin{pmatrix} | & & | \\ \mathbf{a}_1 & \cdots & \mathbf{a}_n \\ | & & | \end{pmatrix}$:

$$\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$$

Equivalently (column combination view):

$$\mathrm{Col}(A) = \{x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n : x_1, \ldots, x_n \in \mathbb{R}\}$$

This is the set of **all linear combinations of columns** of A.

Language: Column space is **constructive** — defined by a **recipe**!

Pick coefficients $x_1, \ldots, x_n$, form the combination.

Column Space is a Subspace

Proposition

$\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

Proof:

(1) Contains $\mathbf{0}$: Take $x = \mathbf{0}$, then $A\mathbf{0} = \mathbf{0} \in \text{Col}(A)$ ✓

(2) Closed under linear combinations:

- Take $\mathbf{b}_1, \mathbf{b}_2 \in \text{Col}(A)$
- Then $\mathbf{b}_1 = Ax_1$ and $\mathbf{b}_2 = Ax_2$ for some x_1, x_2
- For any α, β:

$$\alpha\mathbf{b}_1 + \beta\mathbf{b}_2 = \alpha Ax_1 + \beta Ax_2 = A(\alpha x_1 + \beta x_2) \in \text{Col}(A)$$

✓ Therefore $\text{Col}(A)$ is a subspace!

Example: Column Space of Rank-One Matrix

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

All columns are multiples of $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\text{Col}(A) = \left\{ t \begin{pmatrix} 2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

This is a **line through origin** in $\mathbb{R}^2$!

Visual: All points on line $y = \frac{1}{2}x$.

Connection to Solvability (Lecture 3 Recall)

Recall from Lecture 3:

$$Ax = b \text{ has a solution} \quad \Longleftrightarrow \quad b \in \mathrm{Col}(A)$$

Why?

Because Ax is a linear combination of columns of A, and $\mathrm{Col}(A)$ is **exactly** the set of all such combinations!

Column space captures which vectors can be produced by A.

Linear Independence and Span

Motivation: Redundant Recipe Vectors

Shinchan's coffee shop has recipe vectors (columns = ingredients):

	☕	🍵	🍲
🫘	2	1	3
🍃	1	2	3
🍋	0	1	1

Question: Is 🍲 redundant?

Notice: 🍲 = ☕ + 🍵 (ingredient-wise)

$$\begin{pmatrix}3\\3\\1\end{pmatrix} = \begin{pmatrix}2\\1\\0\end{pmatrix} + \begin{pmatrix}1\\2\\1\end{pmatrix}$$

Yes! We can remove 🍲 and still make everything.

This is **linear dependence**.

The Real Question: Minimum Generators

Motivation for Linear Independence:

Problem: Given recipe vectors $v_1, v_2, \ldots, v_k$:

- Can we write one using the others? (redundancy)
- Can we **remove** some and still generate everything?
- What's the **minimum set** needed?

Goal: Find vectors with **no redundancy** — each contributes something **new**.

This concept is called **linear independence**.

Definition: Linear Independence

Definition (Linearly Independent)

Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are **linearly independent** if:

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \quad \Rightarrow \quad c_1 = \cdots = c_k = 0$$

Intuition: The only way to combine them to get zero is the trivial way (all coefficients zero).

Contrapositive interpretation: No vector can be written as a combination of the others.

Definition (Linearly Dependent)

If there exists a **non-trivial** combination giving zero:

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \quad \text{with some } c_i \neq 0$$

Why This Definition Captures Redundancy

If **dependent**, say $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$ with $c_1 \neq 0$:

$$c_1\mathbf{v}_1 = -c_2\mathbf{v}_2 - \cdots - c_k\mathbf{v}_k$$

$$\mathbf{v}_1 = -\frac{c_2}{c_1}\mathbf{v}_2 - \cdots - \frac{c_k}{c_1}\mathbf{v}_k$$

$\mathbf{v}_1$ can be written using the others — **redundant**!

If **independent**, no such representation exists — **no redundancy**.

Example: Testing Independence

Are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ linearly independent?

Test: Check $c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Only solution: $c_1 = 0, c_2 = 0$ ✓

Conclusion: Linearly independent!

Example: Detecting Dependence

Are $\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ 3 \end{pmatrix}$ linearly independent?

Observe: Column 2 = 2·Column 1, Column 3 = 3·Column 1

Try $c_1 = 3, c_2 = 0, c_3 = -1$:

$$3\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 4 \\ 2 \end{pmatrix} - 1\begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Non-trivial solution exists!

Conclusion: Linearly **dependent**!

Definition: Span

Definition (Span)

The **span** of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ is the set of all their linear combinations:

$$\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_1, \ldots, c_k \in \mathbb{R}\}$$

We say $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ **spans** W if $\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = W$.

Connection to column space:

$$\text{Col}(A) = \text{span}\{\text{columns of } A\}$$

They're the same thing!

Basis: Minimal Spanning Sets

Definition: Basis

Definition (Basis)

A **basis** for a subspace W is a list of vectors that:

1. **Spans** W (every vector in W can be written as their combination)
2. Is **linearly independent** (no redundancy)

Equivalently: A basis is a **minimal spanning set** or a **maximal independent set**.

Key property: A basis gives **unique coordinates**!

If $\{\mathbf{e}_1, \ldots, \mathbf{e}_r\}$ is a basis, then **every** $\mathbf{v} \in W$ can be written **uniquely** as:

$$\mathbf{v} = c_1\mathbf{e}_1 + \cdots + c_r\mathbf{e}_r$$

Why Basis Gives Unique Coordinates

Existence: Spanning ensures you can write $\mathbf{v}$ as a combination.

Uniqueness: Suppose two representations:

$$\mathbf{v} = c_1\mathbf{e}_1 + \cdots + c_r\mathbf{e}_r = d_1\mathbf{e}_1 + \cdots + d_r\mathbf{e}_r$$

Then:

$$(c_1 - d_1)\mathbf{e}_1 + \cdots + (c_r - d_r)\mathbf{e}_r = \mathbf{0}$$

By linear independence, all $c_i - d_i = 0$, so $c_i = d_i$ for all i. ✓

Basis = efficient coordinate system with no redundancy!

Example: Standard Basis for $\mathbb{R}^3$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Check independence:

$$c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + c_3\mathbf{e}_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \mathbf{0}$$

Only if all $c_i = 0$ ✓

Check spanning: Any $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$ ✓

Conclusion: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for $\mathbb{R}^3$!

Example: Another Basis for $\mathbb{R}^2$

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check independence: From $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$:

$$c_1 + c_2 = 0, \quad c_1 - c_2 = 0$$

Add: $2c_1 = 0 \Rightarrow c_1 = 0 \Rightarrow c_2 = 0$ ✓

Check spanning: For any $\begin{pmatrix} x \\ y \end{pmatrix}$, solve:

$$c_1 + c_2 = x, \quad c_1 - c_2 = y$$

Solution: $c_1 = \frac{x+y}{2}, c_2 = \frac{x-y}{2}$ (always exists!) ✓

Conclusion: $\{\mathbf{v}_1, \mathbf{v}_2\}$ is another basis for $\mathbb{R}^2$!

Left Cancellation Property

Matrix Form of Independence

Collect vectors $v_1, \ldots, v_k$ as columns of matrix U:

$$U = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & & | \end{pmatrix}$$

Linear independence says:

$$U \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = \mathbf{0}$$

In other words: $Uc = \mathbf{0} \Rightarrow c = \mathbf{0}$

Left Cancellation Property — Setup

Proposition (Left Cancellation)

If U has linearly independent columns, then:

$$UP = UQ \quad \Rightarrow \quad P = Q$$

Let's write out what each matrix looks like:

$$U = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \\ | & | & & | \end{pmatrix}$$

$$P = \begin{pmatrix} | & | & & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & & | \end{pmatrix}, \quad Q = \begin{pmatrix} | & | & & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_m \\ | & | & & | \end{pmatrix}$$

U is $n \times k$, P and Q are $k \times m$.

Left Cancellation — What Does UP Mean?

$$UP = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \\ | & | & & | \end{pmatrix} \begin{pmatrix} | & | & & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & & | \end{pmatrix}$$

Column j of UP is:

$$U\mathbf{p}_j = p_{1j}\mathbf{u}_1 + p_{2j}\mathbf{u}_2 + \cdots + p_{kj}\mathbf{u}_k$$

This is a **linear combination of U's columns** with coefficients from $\mathbf{p}_j$!

So:

$$UP = \begin{pmatrix} | & | & & | \\ U\mathbf{p}_1 & U\mathbf{p}_2 & \cdots & U\mathbf{p}_m \\ | & | & & | \end{pmatrix}$$

Left Cancellation — Column-by-Column

Similarly:

$$UQ = \begin{pmatrix} | & | & & | \\ U\mathbf{q}_1 & U\mathbf{q}_2 & \cdots & U\mathbf{q}_m \\ | & | & & | \end{pmatrix}$$

If $UP = UQ$, then **column-by-column**:

$$U\mathbf{p}_j = U\mathbf{q}_j \quad \text{for each } j = 1, 2, \ldots, m$$

Rearranging:

$$U\mathbf{p}_j - U\mathbf{q}_j = \mathbf{0}$$

$$U(\mathbf{p}_j - \mathbf{q}_j) = \mathbf{0}$$

Left Cancellation — Using Independence

For each column j, we have:

$$U(\mathbf{p}_j - \mathbf{q}_j) = \mathbf{0}$$

Expanding:

$$(\mathbf{p}_j - \mathbf{q}_j) = \begin{pmatrix} p_{1j} - q_{1j} \\ p_{2j} - q_{2j} \\ \vdots \\ p_{kj} - q_{kj} \end{pmatrix}$$

$$U(\mathbf{p}_j - \mathbf{q}_j) = (p_{1j} - q_{1j})\mathbf{u}_1 + (p_{2j} - q_{2j})\mathbf{u}_2 + \cdots + (p_{kj} - q_{kj})\mathbf{u}_k = \mathbf{0}$$

Since $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ are **linearly independent**:

$$p_{1j} - q_{1j} = 0, \quad p_{2j} - q_{2j} = 0, \quad \ldots, \quad p_{kj} - q_{kj} = 0$$

Left Cancellation — Conclusion

Therefore, for each j:

$$\mathbf{p}_j = \mathbf{q}_j$$

Since this holds for **all columns** $j = 1, 2, \ldots, m$:

$$P = \begin{pmatrix} | & | & & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_m \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_m \\ | & | & & | \end{pmatrix} = Q$$

$$\boxed{P = Q}$$

$\square$

Key step: Independence of U's columns forces each coefficient difference to be zero.

Why "Left" Cancellation?

The matrix U appears on the **left** in $UP = UQ$, and we can "cancel" it.

Contrast: If columns are **dependent**, we **cannot** cancel!

Example:

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

But $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0.5 \end{pmatrix}$!

Cancellation **fails** because columns are dependent.

Consequence: Unique Representation

If $\{v_1, \ldots, v_k\}$ are linearly independent and:

$$w = c_1 v_1 + \cdots + c_k v_k$$

then the coefficients $c_1, \ldots, c_k$ are **unique**.

Proof: If also $w = d_1 v_1 + \cdots + d_k v_k$, then in matrix form:

$$U \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = U \begin{pmatrix} d_1 \\ \vdots \\ d_k \end{pmatrix}$$

By left cancellation: $c_i = d_i$ for all i. $\square$

Cross-Filling Produces a Basis

Recall: Cross-Filling (Lecture 3)

Cross-filling decomposes:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where:

- Each $R_i = \mathbf{u}_i \mathbf{v}_i^T$ is rank-one
- $U = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_r \\ | & & | \end{pmatrix}$ collects the column vectors
- $V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ & \vdots & \\ - & \mathbf{v}_r^T & - \end{pmatrix}$ collects the row vectors

Key question: What is special about U's columns?

Property 1: U's Columns Are Independent

Proposition

Let $A = UV$ from cross-filling. Then the columns of U are **linearly independent**.

Proof idea: Use the **pivot structure**!

Each $\mathbf{u}_i$ came from a rank-one piece with pivot in some row p_i.

After peeling $R_1, \ldots, R_{i-1}$, rows $p_1, \ldots, p_{i-1}$ of the remainder are **zero**.

This creates a **triangular pattern** that forces independence.

Concrete Example: Rank-3 with Off-Diagonal Pivots

Goal: Show cross-filling with **pivots NOT on diagonal**, using **3 colors**.

Let's do cross-filling on a rank-3 matrix:

$$A = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix}$$

Pivots will be at:

- Pivot 1: Row 1, Col 2 — Red
- Pivot 2: Row 2, Col 3 — Blue
- Pivot 3: Row 3, Col 1 — Green

Notice: NOT (1,1), (2,2), (3,3) — completely **off diagonal**!

Step 1: First Pivot (Row 1, Col 2) — Red

Pick pivot at position **(row 1, column 2)**, value 6:

$$\begin{array}{r c l} \textbf{Row 1} & \begin{pmatrix} 0 & \boxed{6} & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix} & \textbf{Pivot 1} \end{array}$$

Extract R_1:

$$R_1 = \underbrace{\begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}}_{\mathbf{u}_1 = \text{column } 2} \cdot \underbrace{\begin{pmatrix} 0 & 1 & \frac{1}{2} \end{pmatrix}}_{\mathbf{v}_1^T = \text{row } 1/6} = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \end{pmatrix}$$

Step 2: After Peeling R_1 — Row 1 Eliminated

Remainder:

$$A_2 = A - R_1 = \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 4 \\ 5 & 1 & 7 \end{pmatrix} - \begin{pmatrix} 0 & 6 & 3 \\ 0 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix}$$

$$\text{Row 1 is ZERO!} \longleftarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \boxed{3} \\ 5 & 0 & \frac{13}{2} \end{pmatrix} \text{Pivot 2}$$

Key: After peeling R_1, **row 1 (the pivot row) became zero**!

New pivot: (row 2, col 3), value 3

Step 3: Extract R_2 — Notice the Zeros!

From remainder $A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix}$, pivot at (row 2, col 3):

Extract R_2:

$$R_2 = \underbrace{\begin{pmatrix} 0 \\ 3 \\ \frac{13}{2} \end{pmatrix}}_{\mathbf{u}_2 = \text{column } 3} \cdot \underbrace{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}}_{\mathbf{v}_2^T = \text{row } 2/3}$$

Observe $\mathbf{u}_2$:

$$\mathbf{u}_2 = \begin{pmatrix} \mathbf{0} \\ 3 \\ \frac{13}{2} \end{pmatrix} \quad \leftarrow \text{First entry is } 0!$$

Why? Because row 1 of A_2 was already zero (eliminated by R_1).

So column 3 of A_2 must have 0 in row 1!

Step 4: After Peeling R_2 — Row 2 Eliminated

Remainder after R_2:

$$A_3 = A_2 - R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 5 & 0 & \frac{13}{2} \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & \frac{13}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$$

$$\begin{array}{r} \textbf{Row 1} \text{ zero} \\ \textbf{Row 2} \text{ zero} \end{array} \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \boxed{5} & 0 & 0 \end{pmatrix} \quad \textbf{Pivot 3}$$

Now: Row 1 AND Row 2 are zero!

Final pivot: (row 3, col 1), value 5

Step 5: Extract R_3 — All Pivots Found!

From remainder $A_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$, pivot at (row 3, col 1):

Extract R_3:

$$R_3 = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}}_{\mathbf{u}_3 = \text{column } 1} \cdot \underbrace{\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}}_{\mathbf{v}_3^T = \text{row } 3/5}$$

Observe $\mathbf{u}_3$:

$$\mathbf{u}_3 = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ 5 \end{pmatrix} \quad \leftarrow \text{First TWO entries are } 0, 0!$$

Why? Because row 1 and row 2 of A_3 were both zero.

Column 1 of A_3 has 0 in row 1, 0 in row 2!

The Perfect Triangular Structure — 3 Colors!

Collected columns from cross-filling:

$$U = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & \frac{13}{2} & 5 \end{pmatrix}$$

$$\begin{array}{r} \textbf{Row 1} \\ \textbf{Row 2} \\ \textbf{Row 3} \end{array} \begin{pmatrix} 6 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & \frac{13}{2} & 5 \end{pmatrix} \begin{array}{l} \rightarrow \textbf{Zero} \text{ (row 1 eliminated)} \\ \rightarrow \textbf{Zero} \text{ (row 2 eliminated)} \\ \end{array}$$

Perfect triangular pattern:

- $\mathbf{u}_2$ has 0 in row 1 (previous pivot row)
- $\mathbf{u}_3$ has 0 in row 1, 0 in row 2 (all previous pivot rows)

Visual Summary: Why Independence? — Backward Substitution

$$U = \begin{pmatrix} 6 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & \frac{13}{2} & 5 \end{pmatrix}$$

Check $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{0}$:

- **Row 1**: $6c_1 + 0c_2 + 0c_3 = 0 \Rightarrow \boxed{c_1 = 0}$
- **Row 2**: $2c_1 + 3c_2 + 0c_3 = 0 + 3c_2 + 0 = 0 \Rightarrow \boxed{c_2 = 0}$
- **Row 3**: $1c_1 + \frac{13}{2}c_2 + 5c_3 = 0 + 0 + 5c_3 = 0 \Rightarrow \boxed{c_3 = 0}$

Triangular pattern (colored zeros!) allows **backward substitution**: solve row 1 $\rightarrow$ row 2 $\rightarrow$ row 3!

General Proof: Pivot Row Method

Let $\mathbf{u}_1, \ldots, \mathbf{u}_r$ be columns from cross-filling, where $\mathbf{u}_i$ has pivot in row p_i.

Key observation: After peeling $R_1, \ldots, R_{i-1}$, rows $p_1, \ldots, p_{i-1}$ of remainder are zero.

Therefore:

$$\mathbf{u}_i[\text{row } p_j] = 0 \quad \text{for all } j < i$$

Check independence: Suppose $c_1\mathbf{u}_1 + \cdots + c_r\mathbf{u}_r = \mathbf{0}$.

Solve backwards:

1. Look at pivot row p_r: Only $\mathbf{u}_r$ has nonzero entry there, so $c_r = 0$
2. Look at pivot row p_{r-1}: Only $\mathbf{u}_{r-1}$ and $\mathbf{u}_r$ could contribute, but $c_r = 0$, so $c_{r-1} = 0$
3. Continue: $c_{r-2} = 0, \ldots, c_1 = 0$

All coefficients zero! ✓

Property 2: $\text{Col}(U) = \text{Col}(A)$

Proposition

If $A = UV$ from cross-filling, then:

$$\text{Col}(U) = \text{Col}(A)$$

Proof (Two inclusions):

Part 1: $\text{Col}(A) \subseteq \text{Col}(U)$

Since $A = UV$:

$$Ax = UVx = U(Vx)$$

Let $y = Vx$, then $Ax = Uy \in \text{Col}(U)$ ✓

Property 2 (continued): The Deeper Direction

Part 2: $\mathrm{Col}(U) \subseteq \mathrm{Col}(A)$

Must show: Every column of U is in $\mathrm{Col}(A)$.

Key idea: Cross-filling **peels columns from A**!

Each $\mathbf{u}_i$ comes from a column of some remainder matrix A_k.

Remainders are built from A's columns, so $\mathbf{u}_i \in \mathrm{Col}(A)$.

Detailed Argument: Induction on Remainders

Cross-filling process:

- Start: $A_1 = A$
- Peel $R_1 = \mathbf{u}_1 \mathbf{v}_1^T$ where $\mathbf{u}_1$ is (scaled) column of A_1
- Remainder: $A_2 = A_1 - R_1$
- Peel $R_2 = \mathbf{u}_2 \mathbf{v}_2^T$ where $\mathbf{u}_2$ is (scaled) column of A_2
- Continue...

Claim: Every column of A_k is in $\text{Col}(A)$.

Proof by induction:

- **Base**: $A_1 = A$ ✓
- **Step**: $A_{k+1} = A_k - \mathbf{u}_k \mathbf{v}_k^T$
  Each column: $(A_k)_j - v_{kj}\mathbf{u}_k$ = combination of A_k's columns
  By induction, both $(A_k)_j$ and $\mathbf{u}_k$ are in $\text{Col}(A)$
  So their combination is too! ✓

Conclusion: Cross-Filling Extracts a Basis

Theorem (Main Result)

If $A = UV$ from cross-filling, then:

The columns of U form a basis for $\mathrm{Col}(A)$!

Proof:

1. ✓ **Linearly independent** (pivot structure)
2. ✓ **Spans** $\mathrm{Col}(A)$ ($\mathrm{Col}(U) = \mathrm{Col}(A)$)

Therefore $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ is a basis! $\square$

Trace and Dimension Well-Definedness

The Dimension Question

Definition (Dimension — Tentative)

If W has a basis with r vectors, we define:

$$\dim(W) = r$$

Problem: What if W has two different bases with **different sizes**?

- Basis 1: $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ (m vectors)
- Basis 2: $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ (n vectors)

Is $m = n$?

Goal: Prove all bases have the same size!

The Tool: Trace

Definition (Trace)

The **trace** of a square matrix A is the sum of diagonal entries:

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

Example:

$$\operatorname{tr}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 + 5 + 9 = 15$$

Key property: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ even if A, B not square!

Subscript Notation: What Does a_{ij} Mean?

$$\begin{array}{cc} & \text{column 2} \\ \text{row 2} & \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & \boxed{a_{22}} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \end{array}$$

a_{ij} **= row** i**, column** j

For a_{22}: first index = row number (2), second index = column number (2).

Visual Proof: $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ — Setup

Consider A is 2×3 and B is 3×2:

$$A_{2\times 3} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \qquad B_{3\times 2} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

First, compute AB (will be 2×2):

Diagonal entries:

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}$$
$$(AB)_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}$$

Visual Proof: $\mathrm{tr}(AB)$

$$\mathrm{tr}(AB) = (AB)_{11} + (AB)_{22}$$

$$\begin{aligned} &= (a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}) \\ &\quad + (a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}) \end{aligned}$$

This is the sum of all **color-matched pairs**.

Visual Proof: Now Compute BA

BA is 3×3:

$$
\overset{B}{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}}
\overset{A}{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}}
$$

Diagonal entries of BA:

$$(BA)_{11} = b_{11}a_{11} + b_{12}a_{21}$$
$$(BA)_{22} = b_{21}a_{12} + b_{22}a_{22}$$
$$(BA)_{33} = b_{31}a_{13} + b_{32}a_{23}$$

Visual Proof: tr(BA)

$$\operatorname{tr}(BA) = (BA)_{11} + (BA)_{22} + (BA)_{33}$$

$$\begin{aligned} &= (b_{11}a_{11} + b_{12}a_{21}) \\ &\quad + (b_{21}a_{12} + b_{22}a_{22}) \\ &\quad + (b_{31}a_{13} + b_{32}a_{23}) \end{aligned}$$

By commutativity of multiplication ($a_{ij}b_{kl} = b_{kl}a_{ij}$):

$$= (a_{11}b_{11} + a_{21}b_{12}) + (a_{12}b_{21} + a_{22}b_{22}) + (a_{13}b_{31} + a_{23}b_{32})$$

This is **exactly the same** sum as tr(AB)!

$$\boxed{\operatorname{tr}(AB) = \operatorname{tr}(BA)}$$

Using Trace to Prove Dimension Well-Defined

Theorem

If W has two bases with m and n vectors, then $m = n$.

Proof strategy:

1. Write one basis in terms of the other (change-of-basis matrices)
2. Use left cancellation: these matrices compose to identity
3. Use trace to count sizes

Proof: Change-of-Basis Matrices

Let:

- Basis 1: $U = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_m \\ | & & | \end{pmatrix}$
- Basis 2: $V = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{pmatrix}$

Since Basis 2 **spans** W, each $\mathbf{u}_j$ can be written using $\mathbf{v}_i$'s:

$$U = VP$$

for some $n \times m$ matrix $P = \begin{pmatrix} | & & | \\ \mathbf{p}_1 & \cdots & \mathbf{p}_m \\ | & & | \end{pmatrix}$.

Column $\mathbf{p}_j$ provides **coefficients** to write $\mathbf{u}_j$ as combination of $\mathbf{v}_i$'s!

Proof: The Reverse Direction

Similarly, since Basis 1 **spans** W:

$$V = UQ$$

for some $m \times n$ matrix $Q = \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{pmatrix}$.

Column $\mathbf{q}_i$ provides **coefficients** to write $\mathbf{v}_i$ as combination of $\mathbf{u}_j$'s!

Proof: Composing the Changes

Substitute $V = UQ$ into $U = VP$:

$$U = VP = (UQ)P = U(QP)$$

Since U has **linearly independent** columns, by **left cancellation**:

$$QP = I_m$$

(an $m \times m$ identity matrix)

Similarly, substitute $U = VP$ into $V = UQ$:

$$V = UQ = (VP)Q = V(PQ)$$

By left cancellation:

$$PQ = I_n$$

Proof: Using Trace

We have:

- $QP = I_m$ (size $m \times m$)
- $PQ = I_n$ (size $n \times n$)

Taking traces:

$$\mathrm{tr}(QP) = \mathrm{tr}(I_m) = m$$

$$\mathrm{tr}(PQ) = \mathrm{tr}(I_n) = n$$

But by the trace property:

$$\mathrm{tr}(QP) = \mathrm{tr}(PQ)$$

Therefore: $m = n$ $\square$

Any two bases have the same size!

Dimension is Now Well-Defined

Definition (Dimension)

The **dimension** of a subspace W is the number of vectors in any basis:

$$\dim(W) = \text{number of vectors in any basis for } W$$

This is **independent of which basis** we choose!

Examples:

- $\dim(\mathbb{R}^n) = n$ (standard basis has n vectors)
- $\dim(\text{Col}(A)) =$ number of vectors in any basis for column space
- $\dim(\{\mathbf{0}\}) = 0$ (zero subspace)

Rank is Well-Defined

Connecting Rank to Dimension

Theorem (Rank = Dimension)

For any matrix A:

$$\text{rank}(A) = \dim(\text{Col}(A))$$

Proof:

From §8, cross-filling produces $A = UV$ where:

- U has r columns
- Columns of U form a **basis** for $\text{Col}(A)$
- Therefore $\dim(\text{Col}(A)) = r$

By Lecture 3 definition:

$$\text{rank}(A) = r \text{ (number of rank-one pieces)}$$

Therefore: $\text{rank}(A) = \dim(\text{Col}(A))$ $\square$

Rank is Well-Defined!

Why? Because:

1. $\text{rank}(A) = \dim(\text{Col}(A))$ (just proved)
2. $\dim(\text{Col}(A))$ is well-defined (all bases have same size)
3. Therefore $\text{rank}(A)$ is well-defined!

Different cross-filling runs (different pivot choices) may give:

- Different U matrices
- Different V matrices
- Different rank-one pieces $R_1, R_2, \ldots$

But they all give:

- **Same number r of pieces** (since $r = \dim(\text{Col}(A))$)
- **Same column space** $\text{Col}(A)$

Example: Independence of Pivot Choices

$$A = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$$

Pivot choice 1 (position $(1,1)$):

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix}$$

One piece $\to$ rank $= 1$

Pivot choice 2 (position $(2,2)$):

$$A = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix}$$

One piece $\to$ rank $= 1$

Both give rank 1 because:

$$\mathrm{Col}(A) = \left\{ t \begin{pmatrix} 2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

Summary

What We Achieved Today

From Algorithm to Geometry:

Lecture 3: Rank was **algorithmic** (count pieces from cross-filling)

- Problem: Different pivot choices might give different counts

Lecture 4: Rank is **geometric**:

$$\boxed{\mathrm{rank}(A) = \dim(\mathrm{Col}(A))}$$

- Solution: Dimension is well-defined (all bases same size)
- Therefore rank is well-defined!

Key insight: Cross-filling **finds a basis** for $\mathrm{Col}(A)$!

The Conceptual Framework

	Descriptive	**Constructive**
Definition	List constraints	Give recipe
Example	Null space: $\{x : Ax = 0\}$	Column space: $\{Ax : x \in \mathbb{R}^n\}$
Strength	Verify membership	Generate members

Basis = minimal spanning set = maximal independent set

Dimension = size of any basis (well-defined!)

Rank = dimension of column space = number of basis vectors from cross-filling

Tools We Developed

1. **Two Languages** for sets: Descriptive $\leftrightarrow$ Constructive
2. **Subspaces**: Flat spaces through origin, closed under linear combinations
3. **Linear Independence**: No redundancy, minimum generators
4. **Basis**: Linearly independent + spans = unique coordinates
5. **Left Cancellation**: $UP = UQ \Rightarrow P = Q$ (for independent U)
6. **Cross-filling produces basis**: Pivot structure $\rightarrow$ independence
7. **Trace property**: $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ (visual proof)
8. **Dimension well-defined**: All bases same size (using trace)
9. **Rank = Dimension**: Geometric meaning of rank